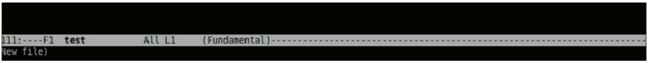

Figure 3: Mode-line displaying Fundamental major mode

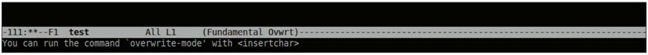

Figure 4: Overwrite minor mode shown on mode-line

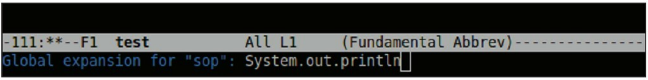

Figure 5: Defining the expansion for "sop"

You can also manually invoke the functions that keystrokes are bound to: if you visit the Key-Index page above, and search for 'C-x C-c', it will bring you to the 'Exiting' link. Click this link. In the Exiting Emacs page, the first text tells you, 'C-x C-c Kill Emacs (save-buffers-kill-terminal)'. Here, the Emacs function that is invoked when you type C-x C-c is named *save-buffers-kill-terminal*. To manually invoke this function, you can type *Alt+x* (also referred to as *Meta-x* or *M-x*) and then type the function name. While this is slower than using the key bindings, it's a very useful feature if you can't remember the binding, or if the function doesn't have a binding (Emacs has a huge number of functions!).

Table 1 has some useful commands and their default key bindings (with *Ctrl* and *Meta* abbreviated as mentioned above) that should help you get started with Emacs.

Now let's look at an important feature of Emacs, which can speed up your work tremendously. I mean the different *modes* in Emacs.

Modes in Emacs

The *vi/vim* editor has two modes: *command* mode and *insertion* mode, which you have to switch between based on whether you are inputting text or commands to *vi*. However, in Emacs, this is not the case; you can run commands by typing a key sequence that's bound to some function, as we saw above—and while not doing that, you can modify the text in the active buffer.

Emacs' *modes* are different; they are *editing modes*, which extend Emacs' capabilities, or change the way some features work, when they are invoked. There are several modes available already in Emacs, for editing each of a certain type or class of data, such as:

- Regular (text) documents
- Source code in a particular computer programming language (C, FORTRAN, Lisp, etc)
- Text formatted in a certain way (outlines, e-mail messages, Usenet articles, character-based illustrations, etc)
- Text with mark-up (like Hyper-Text Markup Language, for example)

There's even a mode for editing non-text (binary) data!

Modes are classified as major or minor. A major mode dictates the main editing behaviour, and is applied only to the active buffer in the current editing session. There can only be one major mode active at a time, though you can switch between different major modes. Minor modes offer some capabilities that are not associated with any particular major mode. Multiple minor modes can be active at a time.

We'll get to some of the available major and minor modes soon, but before that: How do you find out the currently active mode? Well, remember the Emacs mode-line bar we mentioned earlier, which is just below the buffer? This bar displays (among other information) your currently active modes, in parentheses, toward the right side of the mode line. Figure 3 shows an example of this.

The major mode that's active by default (as shown in Figure 3) is Fundamental mode. The simplest of all Emacs modes, it has the fewest special key bindings and settings. Here, the *Tab* key is not bound to a function, and works as you'd expect it to (it inserts a tab into the buffer content). (To see some of the functions the *Tab* key invokes in other modes, search the Key-Index page for 'TAB: Completion Commands' and view the next few entries.)

Here's an example of invoking a minor mode: let's suppose you want to overwrite text that's already entered in the buffer—then you want to go into the *overwrite* minor mode. The function to invoke this is 'overwrite-mode'. To activate this minor mode, either press the bound key, 'Ins' (Insert), or type *Alt+x overwrite-mode*. The mode-line bar changes to show the new minor mode, as shown in Figure 4.